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José Alonso Solís-Lemus

Current position

2019-Present **Post-doctoral Research Associate**, *Developing image processing and computer vision algorithms for the analysis of cardiac images.*, Cardiac Electro-Mechanics Research Group, King's College London.
Prof. Steven Niederer

Education

2015–2019 **Ph.D., Biomedical Engineering**, *City, University of London*, collaborating with Randall Centre for Cell & Molecular Biophysics, Kings College London.

2007–2013 **B.Sc., Applied Mathematics (1st class, Hons)**, *Instituto Tecnológico Autónomo de México (ITAM)*, Mexico City.

2007–2013 **B.Eng., Telematics Engineering (1st class, Hons)**, *Instituto Tecnológico Autónomo de México (ITAM)*, Mexico City.

PhD Dissertation

Title *Segmentation and shape tracking of overlapping macrophages in fluorescent microscopy images.* Supervised by Dr Constantino Carlos Reyes-Aldasoro

Abstract The thesis investigates the complex movement, shapes and overlapping of migrating objects, with special emphasis on images of fluorescently labelled macrophages.

Selected Publications

2023 **Model Reproducibility Study on Left Atrial Fibres**, ***Solis-Lemus***, ···, *Roney and Niederer*, arXiv:2301.06998 [physics.med-ph], (*preprint*).

2021 **Predicting atrial fibrillation recurrence by combining population data and virtual cohorts of patient-specific left atrial models**, *Roney, Sim, Yu, Beach, Mehta,* ***Solis-Lemus***, ···, *and Niederer*, Circulation. Arrhythmia and electrophysiology.

2020 **A simulated single ventilator/dual patient ventilation strategy for acute respiratory distress syndrome during the COVID-19 pandemic**, ***Solís-Lemus***,*Costar, Doorly, Kerrigan, Kennedy, Tait, Niederer, Vincent and Williams*, Royal Society Open Science.

2020 **CemrgApp: An interactive medical imaging application with image processing, computer vision, and machine learning toolkits for cardiovascular research**, *Razeghi,****Solís-Lemus***, ···, *and Niederer*, SoftwareX: ScienceDirect.

2020 **Comparative study of contact repulsion in control and mutant macrophages using a novel interaction detection**, ***Solís-Lemus***, *Sánchez-Sánchez, Marcotti, Burki, Stramer and Reyes-Aldasoro*, Journal of Imaging.

Conference papers

2020 **Software Framework to Quantify Pulmonary Vein Isolation Atrium Scar Tissue**, ***Solís-Lemus***, *Razeghi, Roney, Sim, Mukherjee, Williams,O'Neill, Niederer*, Computing in Cardiology, Rimini, Italy.

2018 **Analysis of interactions of migrating macrophages**, ***Solís-Lemus***, *Stramer, Slabaugh and Reyes-Aldasoro*, Medical Image Understanding and Analysis, Southampton, England, CCIS, Springer.

2018 **Shape Analysis and Tracking of Migrating Macrophages**, ***Solís-Lemus***, *Stramer, Slabaugh and Reyes-Aldasoro*, IEEE Symposium on Biomedical Imaging, WA, US.

Fellowships and Awards

2018	Associate Fellow	*Higher Education Academy (HEA)*
2017	City Graduate School **Conference Travel Bursary**	*University of London*
2015	PhD Studentship	*SMCSE. City, University of London*
2014	Ex-ITAM Research Award in Technology and Engineering	*ITAM*
2012/13	Engineering Students Academic Excellence Award	*ITAM*

Academic Activities

2018 - 2019 **Sessional Tutor**, *Foundation Mathematics / Statistics*, Kaplan International College London, Dr Radmila Topalovic.

2016 - 2018 **Graduate Teaching Assistant**, *Medical Imaging Modalities (BEng) / Medical Imaging (MEng) / Biosignal and Image Processing (BEng)*, City, University of London, Dr Constantino Carlos Reyes-Aldasoro.

Other Professional Experience

2014 **Software Developer**, *Developer specialising in Java and working on network monitoring*, Eficiencia Informativa (transl. *Information Efficiency*), Carlos Fuentes.

2014 **Consulting - Research project assistant**, *Feasibility study for the Puebla ITS Technologies Project with the assessing of the deployment and integration of intelligent transportation system technologies into the Bus Rapid Transit System*, ITAM, Mexico, Dr Andre Possani Espinosa.

Voluntary

2017-2019 **Science writer**, *Student Contribution*, Institute of Physics Publishing, Bristol.

Software development

Since 2020 **CemrgApp**, *An Interactive Medical Imaging Platform with Image Processing and Computer Vision Toolkits for Cardiovascular Research.* , `https://github.com/CemrgAppDevelopers/CemrgApp`.

2020 **Split-Vent**, MATLAB® *package for simulation of a dual-patient/single ventilator strategy.*, `https://github.com/splitvent/splitvent`.

2018 **PhagoSight/Macrosight**, MATLAB® *packages for segmentation and tracking of fluorescent microscopy time-lapse sequences*, Collaboration with CC Reyes-Aldasoro, `https://github.com/phagosight/phagosight`, `https://github.com/alonsoJASL/macrosight`.

Computer skills

Programming	`C++, Python, Matlab, Java, Julia, bash`
Imaging	3D Slicer, ICY, ImageJ, MITK, ITK, VTK
Statistics	`R, SAS, SPSS`